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Letteratura, narratività e composizione nell'era dell'intelligenza artificiale

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Literature, Narrativity and Composition in the age of Artificial Intelligence

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> "The essence of technology is by no means anything technological" – Martin Heidegger
> "No computer has ever been designed that is ever aware of what it's doing; but most of the time, we aren't either." – Marvin Minsky

1 The act of creating something anew has fascinated man ever since the early days of evolution, a quest for curiosity and creation that has moulded man into the beings that they are. The evolutionary transition spurred by the brain's ability to bend, break and blend various elements into newer relations has led to the proliferation of cultural arts and literature[1]. However, for man to build something crudely and analogically similar to that of human brains capable of producing literature (i.e. in the form of AI programs composing literary works), has taken significantly less time than the evolution of the human brain itself. The problem however lies not so much with the capacity of machines to tell stories but with the quality and clarity of their execution. Although AI programs with efficient hardware can process and spurt out text or data at an unparalleled speed, the actual narrative and storytelling output fall significantly short with respect to human-produced literature in terms of clarity, conciseness, and coherence on both semantic and syntactic levels.

2 Artificial Intelligent structures extract, combine and analyse data through machine learning techniques on a continuous basis, employing neural networks and other models that serve their huge databases. Thus, in the age of information whilst processing a lot of information in a small amount of time, these structures truly champion the cause of optimized situations and circumstances by asking less of human effort. While one is surfing the internet through a particular traceable ID, such as a GPS

location or IP address, the user's data is often further fed through content-suggesting algorithms by Big Data companies which in turn personalize one's own web content and give information for an individual's own discourse and ideology. This arrangement shows us what we want to see, but one must always remember the error percentage of algorithms and the predominant availability of 'Free Will' and Sartrean 'choice'[2] that a human being has (however little it be). Discourses in cyberspace are often channelized through the postmodern fetish for 'intensity', the mass distribution of 'viral posts' and meme culture being one such example. Narrativity and narratorial experiences in the contemporary age are not totally independent acts but already deeply embedded in and governed by the web and its algorithmic structures of simulation programs, which further render the reality as a simulation phenomenon.[3] But what happens when machines become narrating beings? In a sense, what significance does it bear when a program, beyond assisting us discursively, starts to perform its own narratives and in turn propound discourses?

3 This paper attempts to address these questions and enquire into instances of narrative composition achieved by A.I. However, for the feasibility of research we shall only enquire into works put out by narrating/composing machines and their implications for cognition, culture and hermeneutics. Composing machines and mechanical composition probably pre-dates the entire episteme of computational composition itself, beginning as early as the Arab astrologer's use of the *Zaijra* and extending to Ramon Llull, Gotfried Leibniz, Jonathan Swift and more modern developments in C.D. structures and L.S.T.M. based Neural Networks[4]. The scope of this paper shall be limited to exploring writing machines specifically rather than delving into the broader realm of mechanical writing. To begin with how narrativity is achieved by machines and their influence on the reader and general culture as a whole, we shall start off by asking the following questions and ruminate on them through the following passages.

- What does it mean to hermeneutically engage with a narrative piece generated by A.I.?
- How do features of cognition and narrativity emerge in trying to make sense of such a text?
- How do schema and frames operate in the reader's cognition when navigating such a text?
- How is narrative coherence achieved and ruptured in such a text?

A.I. and literary composition

4 With the advent of the World Wide Web in the age of Industry 4.0, the praxes of literature and literary activity have radically mutated while permeating across all media forms. The strict adherence to traditional genres and formats has broken down and wide-ranging literary forms, such as interactive fiction or virtual story worlds, have emerged in their place. Literature and literary narratives have been engaging in a direct discourse with A.I. structures from the very dawn of artificial intelligence, the huge database of literature being a perfect and inexhaustible pasture for the experimental grazing of AI programs.

5 There are instances in the history of A.I. in which it has dealt directly with issues of text composition, storytelling and narrativity. Let us take a look at some of them.

- In 1975, Rumelhart[5] proposed the first computational approaches to storytelling, postulating that stories have an internal structure similar to sentences. He broke down the story structure into Setting + Episode, and episodes in turn to Event + Reaction, and Reaction into Internal Response + Overt Response, and Overt Response into Action / Attempt. The goal was to analyse the components of a story and decipher its 'story-ness' and point out that "not all coherent texts are stories". This approach, which can be seen as a systemization of Propp's (1968[6]) proposal for a morphology of folktales, has been followed by a number of researchers (for instance, Mandler and Johnson (1977), Thorndyke (1977).

- Because of their concern with structure rather than content, story grammarians have been criticized as having very little to offer to machine understanding of texts. Another important theory is that of Conceptual Dependency (CD structures), which was developed at Yale University by Roger Carl Schank.[7] In this model, the meaning of individual sentences are represented by CD structures, which consist of concepts and the relationships between them. All information and contextual knowledge implicit in a sentence are laid out explicitly within the CD structure associated with that sentence. However, although CD structures work independently of language because they break down the sentence into its theme and actions, they cannot be applied to complex sentences without their contexts being specified.
- Lehnert's research[8] sought to represent stories as "affect state graphs," which represent mental states and positive and negative events that are causally linked by things like "actualization," "termination," "equivalence," and "motivation."[9]

6 The structures discussed above could however generally process only children's stories and simple sentences. Although BORIS (Behavioral Observation Research Interactive Software) is thought to be among the most "sophisticated understanding programs yet developed," it is only capable of grasping very basic stories on a limited range of subjects, which means that the development of programs capable of understanding *literary* texts is likely still a long way off.[10]

7 Recent research in A.I. and deep learning have developed surprising results in literary and creative activities through neural networking of data and variables. Google has been collaborating with Stanford University to improve the natural language of machines, introducing Google Brain to approximately 11,000 novels.[11] The software is first tasked with understanding the variance of human languages and then given two sentences, out of which the machine composes several poems of its own. Another instance is the project of the Chinese publishing house Cheers Publishing, which put out a collection of poems written by the program Microsoft Little Ice (*Xiaoice*)[12], in which the machine took in more than 500 sonnets and created 10,000 poems 139 of which were published. Another significant A.I. development is a website titled deepbeat.org that generates rap lyrics out of a huge database of rap songs, allowing users to suggest an initial beat or line or even letting the A.I. construct it from beginning to end. The program has recently developed its own voice, which although a bit shaky is quite a nice achievement for an A.I. program. A.I. has also had a breakthrough in screenwriting. orchestrated by scientist Ross Goodwin and his team, which created a Long Short-Term Memory (LSTM) recurrent neural network that renamed itself Benjamin during an interview. It went on to produce a short screenplay titled "Sunspring,"[13] which was made into a film directed by Oscar Sharp that went on to be selected as one of the 10 best short films at the festival Sci-Fi-London. There have also been attempts to produce A.I. novels through deep neural networks, as for example when an A.I. program in 2018 was sent on a road trip to emulate Jack Kerouac: The program/writer was equipped with a microphone, GPS, a camera hooked to a laptop, and a lot of linear algebra. The project was designed by Ross Goodwin and produced the text "1 The Road,"[14] which is currently marketed as the first novel written by A.I. The program ingested 60 million words (approximately 360mb) for its

processing. However, it is quite a frustrating read for its surreal use of language, which skews often into gibberish. Still the program achieves sentence coherence, which was a big step forward from the time of CD structures.

8 A recent research project by Open A.I.'s GPT-3[15] and GPT-4, which can be partly accessed via talktotransformer.com (GPT-3), consists of a well-trained neural network that is capable of generating simulated stories or even 'deep fakes' by taking an initial cue from the user and completing a convincing line or paragraph for them. In addition to producing direct compositions in the past few decades, A.I. has also served the purposes of literary analysis by mining big data sets pertaining to usage. Programs such as "Witch Hunter"[16] (a geo-semantic map that traces the hotspots of witchcraft activity in Denmark by mining 30,000 stories and their geographical contours) and "TimeML" (a tool that charts the spatio-temporal distinctions and pace of a narrative) use advanced machine learning techniques to bring "distant reading" techniques into a utilitarian forefront. Douglas B. Lenat's brainchild "CYC"[17] is probably the oldest existing artificial intelligence program that seeks to lay down a vast epistemological framework of basic human behaviour and common-sense. The approaches of cognitive linguistics and cognitive neuropsychology with their stress on the computational prospects of the mind in general and its literary features in particular, can further alleviate current shortcomings of A.I. research in narrative generation. As far as the future of narrative-competent A.I. goes, the so-called trans-epistemic traffic between the study of literary processes in the human mind and the conceptual challenges in attempting to represent them artificially are mutually productive and sustainable.

Cognitive-Literary Analysis of A.I. compositions

9 So far we have discussed some landmark moments of A.I.'s involvement in literary creation and criticism without, however, engaging in a heuristic analysis of the corpus produced. The following section shall try to address the qualitative aspects of an A.I. generated text and examine their narrative details. To account for the comprehensibility of texts composed by an A.I. program, we resort to a cognitive literary analysis of the generated works. In order to account for narrative coherence, continuity and contingency, we shall adopt the methodological premises and framework of cognitive semiotics[18] and cognitive poetics[19]/narratology[20]. By analysing the texts in terms of syntactic regularity, lexical similarity, metaphorical mappings, mental spaces[21] and hermeneutical potency; we enquire into the frameworks of narrative coherence as achieved, exhibited or implied. For the sake of feasibility and the scope of this paper, we shall limit our idea of the reader as essentially a reader/interpreter—one who engages in a dynamic cognitive exercise with the text—and not go into the subtleties of what it means to be a reader (human or otherwise). The following texts have been chosen as some of the unique and experimental works of literature produced by powerful fourth generation A.I. with state-of-the-art technological advancements. We shall begin with a cognitive-poetic analysis of Xiaoice's poetry and gradually move into further intricacies of narrativity by examining "1 The Road."

The Sunshine Lost Windows by Xiaoice

10 Let us begin with one of the most popular poems composed by Xiaoice:

> The rain is blowing through the sea
> A bird in the sky
> A night of light and calm
> Sunlight
> Now in the sky
> Cool heart
> The savage north wind
> When I found a new world...

11 The poem begins with an initial frame over a seascape, one that invokes either scripts or frames of gentle rain. However, the verb "blowing" is conceptually laden with the domain of "force" and is more likely to evoke the forceful effect of a storm. Following this sight, our schema advances with the bird in the sky, a symbol of presence and motion projected via a simple figure/ground relation. In the second line, are we to understand that the bird occupies the same frame as the first line or are we to imagine a different frame (against a backdrop of clearer skies, for example)? The answer isn't narratively explicit or coherent. Whatever our interpretative choice in the second line, the third line violently introduces a newer frame of the night being "light and calm", which is quite ironic following the rapid shifts of frames, moods and emotions. The fourth line, consisting of one word—"Sunlight," brings an entirely different state to the foreground. A singular noun occupying a single line for the sake of poetic effect is juxtaposed to the frame of night developed previously. In its contrastive effect of the night and day sky frames, the schema develops with a rapid influx of luminosity that constitutes both a brief pause and extends the narrative progression. As one navigates through these instantaneous changes in temporality, a conceptual insight on the rapid nature of frame dynamics itself emerges. The next lines direct us back to the schema of the sky, pushing us to imagine "cool heart" as a metaphor for an internal emotional state. Furthermore, the appearance of "cool heart" just after "...in the sky" activates the Conceptual Metaphor[22] of GOOD IS UP, BAD IS DOWN wherein the sky by default projects the ontic of the "heart" as benign and sufficiently elevated. In the next line, we continue with our "cool heart' elevated and fluttering in the "savage north wind," or we alternatively end the frame of the cool heart in the sky. Instead, "the savage north wind" now almost blows into the next frame with a glimpse of self-other (re-)awakening and a discovery in which the "new world" is found. It is interesting to note the presence of an "I" in the final line as it instantaneously puts the entire poem into the deictic axis[23] of the implied author or even the reader (who is mentalizing in the process). One who has witnessed or even dreamt perhaps, the rapid frame shifts on an enactive[24] and embodied[25] level and emerges teleologically with the discovery of a new world.

12 The hermeneutical[26] possibilities, though various, remain limited and disorganized in light of the incoherent rhetoric and lexical incongruity. However, the reader, as a result of his or her lived-experience and being-in-the-world-ness and the remarkable efforts of the narrativizing brain,[27] coherently arranges the information into a situated and acted-out narrative. The power of narrative is enough to coax the reader into pleasurably gliding through the text without realizing the nature of its authorship. Indeed, it doesn't read as being as experimental as e.e. cummings or as hip as Ginsberg.

However, the point isn't to *emulate* but rather to *create* out of scratch with the coherence of a presentable text. It is this task which Xiaoice achieves quite well, despite all the shortcomings with regard to robustness of L.S.T.M. processing. Of the 117 poems published in *The Sunshine Lost Windows*, some poems remain unique in their iconicity and diction while imparting a cogent sense of progression to the narrative. Consider the following lines composed by Xiaoice:

> Cities are too ashamed to face the countryside,
> Let us compare beautiful hearts.
> And concentrate a scenery,
> Like swaying with the wind[28]

13 The poem begins with an assertive statement of comparison across the nature-culture binary apparatus. Cities with all their tentacular and network-based complexities are ashamed to come face-to-face with the simpler countryside, an image that evokes man's eternal guilt for not taking good enough care of its all-benevolent mother (who seems to be ailing now). The second line comes almost as a demand asking us to compare "beautiful hearts," beings that are not unlike each other. Following this, the generic frame of "a scenery" calls conceptually on ideas ranging from physical cities to biological sensations of shame to countryside to human metaphors of heart. The poem ends with the metaphor of "swaying in the wind"- a frame captures the characteristics of fleeting . The final line compliments and completes the frames built up in the first two lines, in drawing itself to the metaphor of swaying as an act of transience-presence and ethereal ephemerality. The program Xiaoice brilliantly crafts the poetic form onto the generated text by either switching each word/sentence holding a significant frame or an accretion within a given frame onto the next line. The switching between different frames and schemas with each syntactic unit (each new line) alludes formally to traditional aspects of verse, like rhythm and rhyme.

14 The cogent expression of the poem however comes across us to tell a narrative, a story; as Paul Ricœur notes, stories are "models for the redescription of the world."[29] Once triggered by the text, the narrativizing mind seeks to reinforce the models of the world alongside it, for as Jerome Bruner notes "the story isn't by itself a model. It is, so to speak, an instantiation of the models we carry in our minds."[30] *The Sunshine Lost Windows* itself is evidence of the rapid developments in our capacity for creating texts through artificial means as well as a harbinger of the potential cultural impact of such texts. Despite having a degree of lexical, semantic, and grammatical incoherency, Xiaoice's poems somehow seem to make sense, given the fact readers can cognize them. It would seem that, in order to make meaning of the narratives composed by A.I. programs, the reader needs to be able to move across different semantic frames, schemas,[31] and domains and link them.

1 THE ROAD by Ross Goodwin

15 Ross Goodwin's phenomenal project seeks to expand the horizons of writing and composing literature, as well as deconstructing and enhancing its processes, experiences and components. *1 The Road* is marketed as the first novel composed by an A.I. program and emulates Jack Kerouac's *On The Road* by actually sending its maker on the road quite literally. The set-up consists of a car mounted with a camera, microphones, a GPS-tracker and a printing machine. Its program runs a sophisticatedly trained recurrent neural network while using LSTM that had ingested a corpus of

approximately 60 million words, comprised of equal parts poetry, science fiction and 'bleak' literature.

16 As Ross Goodwin aptly describes the project:

> Fueled by exploding gasoline, artificial neurons firing raw data through burning cores, not so much written as scorched onto each roll of thermal paper — by phantom authors, a mechanized pen, drivers at the wheel, or perhaps the road itself — from NYC to NOLA: 1 the Road is a book written using a car as a pen.[32]

17 On encountering the novel one shouldn't be surprised at the incongruous and experimental narrative framings—spurting out conversations, map co-ordinates and poetry in a singular conversational strand. Being the first project of its kind, the message is as important and clear as the medium.[33] Let us consider the following excerpt from the novel for a detailed analysis of narrative coherency and sense-making:

> Three seconds after midnight. Coca-Cola factory, Montgomery. A building in Montgomery to his father's study of this town in the same room where the band was being sent off to the police car. The time was one minute past midnight. But he was the only one who had to sit on his way back. The time was one minute after midnight and the wind was still standing on the counter and the little patch of straw was still still and the street was open.[34]

18 On encountering the above lines, one is certainly struck by the eerie descriptions and surrealistic offshoots and projections they espouse. However on a closer reading, the passage seems to be pervaded by a notion of proper spatio-temporal ordering alongside elements of narrative that engages surrealistically with the background progression. It starts off with registering the time, then the location and attaches a piece of narrative related to either of the above pre-conditionals. The narrative that emerges often suffers from lexical incoherence and thus, despite achieving new blends[35] in the reader's mind, fails to adequately maintain the same space for lack of explicit domain detailing and threading. The lines above, however, display a lucid adherence to coherent progression and could at times even be said to be poetic or surrealist, at least compared other A.I. produced texts as yet in circulation. The text's strikingly harmonious resolution of incoherent semantic constructions, which produces a surreal effect, is a result of its uniquely blended spaces and metaphors (de)constructed in the reader's cognitive exercise. In the last line of the excerpt, we stumble upon an exquisite personification of the wind standing still on the counter alongside the little patch of straw, an image that squeezes the most miscellaneous agents into a single frame with laudable syntactic coherency. The text exhibits its novelistic form by sticking to syntactic continuity even within the rapid shifting or accretion of frames. A single full stop often separates one frame from another. However, the episodic progression informs the general schematic development of the text. In noting the spatio-temporal coordinates within the generative narrative piece, *1 The Road* can come across as an avant-garde novel doing a pastiche of the travelogue or the journal-entry genre.

19 *1 The Road* also provides an avenue for an enactivist and embodied account of the narrative as established by Goodwin's program. Goodwin's attempt to reinforce learning and, in turn, composition by using various data input sources (the camera, microphone, the GPS tracker), with the end goal of providing a more coherent sense of narrative, highlights the importance of these stimuli in the overall process of narrative composition. It is the perception-action chains of the vehicle apparatus involved in cruising through the American landscape that is responsible for the narrative structure

of the text. However, to ascribe direct embodied and enactive processes to something so mechanical yet 'organic' might be a misguided task, but one that needn't be discarded outright. Goodwin's program blurts out letters, words and sentences based on a statistical association processed through the neural network without providing a perpetuated sense of narrative coherence or association. Although the articulated prose sometimes comes off as strikingly poetic and surrealistic in guise, it is often because of the mental spaces evoked (incompletely, leading in turn to possible worlds[36]) are offered up to the reader's cognitive architecture, which fills in, makes sense, and understands.

20 Goodwin however feels responsible for the authorship of the work despite the fact that it is composed quite independently of him by the neural network system. Perhaps the sense of authorship stems from the fact that, despite the text generator's independence, it still formulates within certain parameters set by the algorithms (which were, of course, designed in turn by Goodwin). Goodwin lays out an ethical statement and stance for the project when he says that the purpose is also to reveal how machines make sentences: "In the future when this text becomes more sophisticated, it's a warning. If you see patterns like this, it may not have been written by a human."[37] By giving a glimpse of what bot-generated content might someday be able to achieve, Goodwin creates a manual to follow in order to avoid fake and manipulative content driven by the biased interests of the few.

Further Implications for Composing A.I. structures: Gaps, Traps and Maps

21 So far, the reader remains elemental in the role of grazing texts in that eternal pasture of hermeneutics that is fenced in only by context and cognitive limitations. It is human cognitive architecture,[38] with its capacity to bend, break, and blend, that accounts for all the sense-making and dis-junctures encountered while narrativizing. In recent years, A.I. and Machine Learning have advanced drastically in technical and computational fields dealing with "memory" and "information." However, literary applications in these fields are still in their infancy, and applications with regard to both literary creation and analysis suffer from a lack of conceptual integrity; for, although A.I. is capable of applying neural networks, it does not *understand* and is not *aware*. Leaving aside the problems of programming unconscious processes or even consciousness itself, the very complex nature of the subjective-poetic mind and its interaction with the work makes it difficult or even impossible to detect and comprehend the working principle of this dynamic. Art is not a direct product of consciousness alone but exists in the complex traffic of an elaborate cognitive and conative framework alongside an enactive encounter with the creator's environment. It exists in the de-territorialising that constitutes the poiesis and praxis of what-is, what-isn't and what-can-be. The creative process is characterized by both a self-awareness that resides in subtle bursts of introspective analysis during the composition of an artwork and in the ultimate loss of that awareness in the work itself. The works of art put forth by A.I. lack the "true artistic genius" that emerges from the confluence of multiple factors, including both *a priori* knowledge and lived experience. While an A.I. program can compose a Dadaist poem along the lines of the process that Tristan Tzara described in which an algorithm takes random words and generates "original" lines,[39] it

would be impossible for a program to compose a poem like "The Wasteland" by T.S. Eliot, which – although it may appear to be an assemblage of juxtapositions – remains a contextually based and conceptually constructed artefact that is hermeneutically stimulating and a remarkable feat of poetic composition.

22 Reading and interpretation are dependent on the receptive value of the text and its cultural uses. Our minds are brilliant composing and comprehending "machines" to begin with, spontaneously capable of dealing with "possible worlds." The hermeneutical stance enjoyed by the reader cannot be qualitatively formalized by mere true-false statements. Perhaps, it is precisely what Aristotle calls *dynamis*[40] (potentiality) that is at work in the cognitive processing of the text by the reader, in the service of the *logos* of reading. Perhaps the key to a coherent narrative cannot be achieved and maintained by a neural network model alone that begins with a prompt and then cues in passages based on the most likely possible arrangements inferred from a probabilistic distribution of the input dataset. There needs to be an underlying context-based and concept-based structuring that begins with a consideration of both a bottom-up and top-down approach working in tandem—a sort of rule-based system that respects the basic schema patterns and frame semantics and organizes them based on the contexts in which they most often appear whilst allowing enough room for non-normativity and deviancy by maintaining a coherent awareness of the entire frame or schema. The work of art is often characterised by its non-normative and non-conformist nature that ushers in the "new" by invoking and deviating from "idealized cognitive models."[41] The diversion from idealized models of narrativity and sense-making in a work of art occur within the limits of what Marie Laure Ryan calls "the principle of minimal departure"[42] that dictates the formation of both mental spaces and possible-worlds. The works produced by A.I. programs are still at a pioneering stage of creation and are liable to not make coherent sense and to use excessive poetic license in ways that do not often conform to the models, so they cannot be expected make coherent and comprehensible "departures." The narratives in the texts generated often pose significant challenges for recognition, coherent reading, and mentalizing. As Nancy M. Ide and Jean Veronis note:

> The overall meaning of a narrative is not reducible to a final and static representation of the events of which it is composed, but is also derived from the successive steps that the reader takes in the process of reading the text.[43]

23 The work of hermeneutics then increasingly involves the construction of new models at every turn, something which is already quite intrinsic to the ontology of hermeneutics itself. However, the level of abstraction inherent in A.I.-generated texts calls for something more from the realm of hermeneutical enquiry. Perhaps, it calls for a dynamic filling-in on the part of the reader's imagination by achieving conceptual blends or metaphors that might not be significant at all but that are nonetheless present. In dealing with the narrative anomalies of an A.I.-generated text, the importance of the reader becomes paramount. Such a stance adheres and attests to the co-dependent productive relations of man and machines about which Ross Goodwin notes that the "generation and interpretation cycle demonstrates the true augmentative capacity of these learning machines."[44]

24 The fields of artificial intelligence and machine learning stem from the mechanical and computational aspects of intelligence, the basic traits of which were inferred by humans via first-hand study and observation of the human mind. Neural computation

is however *sui generis*[45] and bears no resemblance to the workings of a digital A.I. program that belongs to altogether different planes of ontology, dynamics and biochemistry. There lies significant differences between the algorithmic structures used to generate a narrative and the human cognitive structures used to generate, infer, and move through the narrative. One of the most fruitful research possibilities that can help to strengthen the praxis of A.I. functionality in terms of text comprehension and composition is to locate and try to emulate similar functional underpinnings within our own brain-mind architecture, since such processes that involve the interplay of blending, bending and breaking are uniquely inherent to the human mind. Research into conceptual integration[46] across semantic domains and the cognitive mechanisms that produce creative compositions can further help us understand how such processes occur and thereby help us to engineer robust composing programs. Deeper research into cognitive science and cognitive literary studies, as well as into the computational aspects of creative cognition, might perhaps help put elucidate a wide range of aspects of narrative framing and comprehension on a logical and pragmatic level. Using brain imaging (via PET, CT-scan, *f*MRI) to analyse subjects performing literary tasks such as interpretation, on the other hand, could further help to test cognitive theories while avoiding the "consistency fallacy."[47] Although the capacity for tracing neural correlates for narrative composition may be at a rudimentary stage, brain imaging studies could, with technological advancements, help to shed light on the undercurrents of narrative processes in the mind. Such research could serve not only to note how such mechanisms are undergone via neuronal and nodal networks but also their further implications with regard to general cognitive processes.

A significant approach that aims at a cognitive view of preliminary narrative formation is that of the "Theory of Mind,"[48] which aims to understand the desires, beliefs and intentions of a person from another perspective. The dualist approach of Theory Theory (TT) and Simulation Theory (ST)[49] together pave the way for capturing the context-based inference and plausibility of human thoughts and causality. It is far too early to predict a programmable version for the Theory of Mind or Enactivism since most of the mind's cognitive and conative architecture remains latent and un-representable, yet with advanced and persistent research such undercurrents should come to light. Should it suffice to say then that an A.I. program will actually "understand" what its correspondent is referring to, in terms of performing a mentalizing action? Perhaps it is too early to entertain the idea of such functionality, for an A.I. program needs to have both a certain level of consciousness and embodied, enactive cognition to participate in acts of "subjective" mentalizing rather than just optimizing with respect to the guidelines initially programmed by/with human intervention. Consciousness remains far from programmable by the very fact that it is not yet detectable and strictly recognizable, on either a local level or across the broad spectrum of mind and neural networks. A recent 2014 study at the University of Wisconsin-Madison posited a surprising hypothesis in grasping/annotating the "presence" of consciousness both empirically and formally: Giulio Tononi postulated that a being's "consciousness" is directly proportional to the amount of "integrated information" it possesses.[50] "Integrated Information Theory" (I.I.T.) accounts quantitatively for how the integration density of a system implies a higher state of synergy, which in turn determines the depth of the system's conscious experience[51]; whilst informational geometry in turn seeks to account for the qualitative experience of consciousness or *qualia*.[52] Such a theory starts from 'experience itself' and notes its essential properties (axioms) in order to identify characteristics within physical systems that can support such axioms (postulates)[53].

26 Perhaps someday in the future we might be able to apply these quantifying measures to physical systems comprising A.I. programs and thus (in principle) be able to determine their relative levels of consciousness. One of the most challenging factors, besides that of programming conscious experiences, is the idea of creativity and imagination as acts that can be meaningfully simulated. However, such a one-time simulation precludes the very possibility of "originality," which allows for both following rules and breaking them and which endows the creative act with its very charisma.

27 Narrative and literary texts incorporate elements and elemental knowledge of vast discursive fields across all epistemes. Thus, to understand, create, or analyse literature, the machine has to have access to information to which even humans do not have direct access. For it isn't possible to either program all the *a priori* knowledge or account for a systematic formulation of all the cultural knowledge acquired post-birth. Machine learning techniques, through the synthetic formulation of the logic of natural systems into algorithms, uses problem-tackling scenarios based upon the data sets on which it was trained. While encountering a new data set (that may be dramatically different from the previous ones), the program can exhibit higher rates of anomaly. The processing of text by humans, on the other hand, is by default subject to multiple interpretations, many contradictory or conflicting, but each given a respectable position in its contextual development. A possible explanation for the act might be our innate ability to simulate, inhabit overlapping "schemas," construct frames, and metaphorically co-relate or bind[54] distant domains while exercising what Mark Turner calls "double-scope blending" with relative ease. This problem might also illuminate the reasons for the wide constructional gap between narrow A.I. and Artificial General Intelligence.

28 Even if there might emerge in the near future an artificial cognitive apparatus that emulates conscious-like behaviour, it will have the same problem with regard to dealing with texts that the homologists have with social analysis in the critique of Raymond Williams—its ideology being easily separable from its lived experience.[55] The system can be programmed without (embodied) contextual relevance, giving rise to a situation labelled as "false consciousness" by Adorno and Lukács.[56] Unless, of course, the program is enactively designed with a corporeal apparatus (like a cyborg) mimicking the cognitive and social processes of humans and the ability to learn heuristically. The act of stimulating the processes of the creative human mind within an algorithm or network is still at a basic and developing stage of research, yet further work should be able to shed more light on the epistemic gaps with regard to advancements in time and technology. However, it is for the very non-comprehensibility of the exact working principles of creative and comprehensive processes that A.I. still struggles to generate its own sensical narratives (which is a reflection of our inability to pin down the same processes). Although we experience and understand a piece of literary narrative with relative ease, we cannot precisely codify the ways in which we "experience" or "understand" them. Perhaps, a wider understanding of general cognitive mechanisms and their relational combinations in presenting sensical narratives can help to address these shortcomings.

29 A.I. and Literature still have a long way to go, and each aspect of their mutual development is augmented by a relentless traffic of cross-disciplinary (ex)changes. In an age of rapid automation and encoded-ness, it is perhaps literature and art that can give one solace from the roller-coaster ride of information and excess. The function of art, however, is not to defy technology or escape the information age; rather it develops new forms with every technological advancement. Art and literature in positing an inexhaustible *noumena*-like ontic, as espoused by Ortega y Gasset,[57] remain ever-refreshing forms that mime the very dynamic nature of the mind, the world, and their complementary transactions. The work of art thus provides a site for not only exploring the world and its relation to the self, but also the shifting relations that (re-)create the semblance of the self while navigating through the world-at-hand, a feature that has been useful for humans and might be for machines as well.

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ABSTRACTS

My paper attempts to briefly trace the interactions between A.I. and literature while noting the various ways in which machine intelligence participates in the generation and comprehension of narrative. Modern research in A.I. and narrative theory remains deeply involved in the pioneering field of narrative generation, while literary works composed by A.I. are scant, facing major issues with regard to comprehensibility and innovation. What are the factors that limit the literary potential of narrative creation by machines? Can such factors be curbed and how? My paper will briefly try and address these questions regarding the shortcomings of narrative generation and processing by A.I. I shall employ a cognitive-narratological framework in assessing several narratives created by A.I. programs, focussing on schema and frame designs and the lack of coherence in generating meaningful utterances. The paper shall also attempt to highlight how such incomprehensible or often dis-coherent compositions are apt to be cognised by human readers. What does this tell us about our own cognising and mentalizing abilities? The paper will address problems of cognition that play a major role in the development of logical and algorithmic structures for the process of writing as a whole and foretell the limitations of writing machines (whether sentient or not). I believe that the various interactive zones between literary narratives and artificially intelligent structures open up an entire new field of discourse that is worthy of exploration—a field in which the text, society and digital systems come together in a state of "libre jeu."

Mon article tente de retracer brièvement les interactions entre A.I. et la littérature tout en notant les différentes manières par lesquelles l'intelligence artificielle participe à de tels actes de génération et de compréhension narrative. La recherche moderne en A.I. et la théorie narrative pointe encore vers le domaine pionnier de la génération narrative dans lequel les œuvres littéraires composées par les IA sont rares, confrontées à des problèmes majeurs de compréhension et d'innovation. Quels sont les facteurs qui conduisent à de tels actes appauvris de création narrative? Comment ces facteurs peuvent-ils être freinés (voire pas du tout)? Mon article tentera brièvement d'aborder les problèmes et les lacunes suivants de la génération et du traitement narratifs par A.I.s. J'emploierai un cadre cognitivo-narratologique pour évaluer les récits élaborés via A.I. programmes, en soulignant spécifiquement les domaines des conceptions de schémas et de cadres et le manque de cohérence qui en résulte dans la génération d'énoncés significatifs. Le document tentera également de parcourir les récits composés par A.I. programmes, en mettant au premier plan comment de telles compositions incompréhensibles ou souvent dis-cohérentes sont souvent connues par le lecteur humain. Qu'est-ce que cela nous informe de nos propres capacités de connaissance et de mentalisation? L'article abordera les problèmes de cognition qui jouent un rôle / obstacle majeur dans le développement de structures logiques et algorithmiques pour le processus d'écriture dans son ensemble, impliquant l'indisponibilité d'une machine à écrire (qu'elle soit sensible ou non) et la disponibilité d'écrits mécaniques. . Je crois que les différentes formes des zones interactives entre récits littéraires et structures intelligentes artificielles ouvrent un tout nouveau champ de discours digne d'un exercice cognitif; où le texte, la société et les systèmes numériques sont mis ensemble dans un état de libre jeu.

Il seguente articolo tenta di tracciare brevemente le interazioni tra I.A. e letteratura, oltre ad analizzare i modi in cui l'intelligenza artificiale partecipa a tali atti di generazione e comprensione narrativa. La moderna ricerca in I.A. e la teoria narrativa puntano al campo pionieristico della produzione letteraria - nel quale le opere composte da I.A. sono ancora scarse - affrontando problemi di comprensibilità e innovazione. Quali sono i fattori che conducono agli attuali poveri risultati di creazione narrativa? Come possono questi fattori essere limitati (se non del tutto eliminati)? L'articolo tenterà di affrontare le attuali carenze dell'I.A. nell'elaborazione narrativa. Utilizzerò un quadro cognitivo-narratologico nella valutazione dei testi generati tramite programmi I.A., focalizzandomi nel design di strutture narrative e nella mancanza di coerenza semantica nel generare enunciati significativi. L'articolo cercherà inoltre di esaminare alcuni testi scritti artificialmente, evidenziando come tali composizioni spesso incoerenti - o talvolta incomprensibili - siano comunque riconoscibili dal lettore umano. Cosa implica tale fatto riguardo le nostre capacità cognitive? L'articolo affronterà in seguito i problemi cognitivi che contribuiscono (o ostacolano) lo sviluppo di strutture logiche e algoritmiche durante il processo di scrittura nel suo insieme, tenendo conto della scarsità di macchine (senzienti o meno) in grado di scrivere autonomamente e dei pochi "testi artificiali" attualmente disponibili. Ritengo che le interazioni tra narrazioni letterarie e strutture intelligenti artificiali aprano un nuovo campo di ricerca cognitiva, nel quale i testi letterari, la società e i sistemi digitali si relazionano in uno stato di "libre jeu".

INDEX

Mots-clés: cognition, cohérence, Intelligence Artificielle, composition
Palabras claves: coherencia, composición, Inteligencia Artificial, narratividad
Keywords: cognition, coherence, Artificial Intelligence, narrativity

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